MINI PROJECT SYNOPSIS

ON

HealthSense AI: AI-Powered Family Health & Smart Food Recommendation System

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1. INTRODUCTION

In modern households, packaged and ultra-processed food products have become ubiquitous staples of daily nutrition. However, interpreting the nutritional content of these items poses severe challenges for typical consumers. Commercial food packaging frequently displays complex chemical additives, cryptic E-numbers, and ambiguous marketing assertions such as *'zero cholesterol'*, *'sugar-free'*, or *'natural'*, which frequently obscure high levels of hidden sodium, trans-fats, or hazardous allergens. In an Indian family setting, this nutritional opacity is compounded by the fact that multi-generational households accommodate individuals with starkly divergent physiological requirements—ranging from elderly grandparents managing Type-2 diabetes and hypertension, to adults with dyslipidemia, and children with severe peanut or dairy allergies.

HealthSense AI is engineered as an intelligent, mobile-based food-health decision support system designed to bridge this critical informational divide. Operating as a hybrid system comprising a high-performance native Android application and a robust Python/Django machine learning microservice, HealthSense AI empowers families to maintain independent health profiles under a consolidated household account. By simply scanning a packaged product's barcode with the device camera, the system retrieves its complete nutritional and ingredient taxonomy, applies clinical deterministic safety rules alongside state-of-the-art machine learning classifiers, and generates personalized, explainable recommendations classified as **Safe**, **Consume with Moderation**, or **Avoid**.

2. PROBLEM STATEMENT

2.1 Working of Existing System

Currently, consumers rely almost exclusively on manual label reading or generic food rating applications (e.g., Yuka, Open Food Facts, MyFitnessPal). These conventional systems suffer from three fundamental architectural limitations:

- **Single-User Bias:** Existing apps assume an isolated, healthy adult user. They cannot evaluate how a single grocery item impacts multiple family members with conflicting medical restrictions simultaneously.
- **Lack of Clinical Determinism:** Generic rating engines compute an unweighted aggregate score (e.g., Nutri-Score A to E). They fail to enforce hard clinical contraindications; for instance, a cereal scoring 'A' due to high fiber may still prove fatal to a family member with celiac disease due to gluten content.
- **Opaque Black-Box Outputs:** Most commercial health apps present arbitrary scores or binary red/green badges without explainable feature attribution, leaving users unaware of the specific biochemical trigger causing the rating.

2.2 Need of New System

There is an acute necessity for an integrated, clinically aware decision-support platform that unites barcode scanning, multi-profile family management, deterministic medical safety filters, and machine learning recommendations with explainable AI. HealthSense AI addresses this gap by ensuring absolute hard-stop protection for severe medical contraindications, followed by granular ML suitability grading and human-readable SHAP-derived explanations, establishing transparency and empowering families to make safe, informed nutritional purchases in real time.

3. SCOPE OF PROPOSED SYSTEM

The scope of HealthSense AI encompasses the full-stack design, engineering, and empirical validation of an end-to-end mobile healthcare decision-support system. The functional boundaries of the system are defined as follows:

• **Multi-Member Family Profiling:** Securely store and manage independent clinical profiles (age, gender, BMI, diagnosed chronic illnesses, severe allergens, dietary preferences) under a single authenticated household account.

• **Sub-Second Barcode Identification:** Integrate Google ML Kit and CameraX on Android to achieve fast (<500ms) on-device 1D/2D barcode scanning across EAN-13, UPC-A, and QR standards.

• **Nutritional Taxonomy & Ingredient Parsing:** Ingest and normalize raw product data from the Open Food Facts global database and USDA FoodData Central, parsing complex multilingual ingredient strings.

• **Two-Tier Hybrid Recommendation Engine:** Enforce a deterministic Layer-1 safety rule engine (absolute hard-stops for 9 major allergen categories and clinical thresholds for diabetes, hypertension, and renal disease), coupled with a Layer-2 pluggable ML classifier (XGBoost, LightGBM, Random Forest).

• **Explainable AI (XAI):** Generate localized feature-attribution summaries via SHAP (SHapley Additive exPlanations) to present plain-language justifications for every recommendation.

• **Offline Client Fallback:** Implement a lightweight client-side HealthRuleEngine within Android to guarantee basic allergen and safety checks even when mobile internet connectivity is completely unavailable.

Operational Boundaries: HealthSense AI is designed specifically for commercially packaged food products bearing standard barcodes. It is strictly scoped as an educational decision-support tool and does not provide formal medical diagnosis or replace consultation with licensed physicians.

4. OBJECTIVES OF PROPOSED SYSTEM

The primary engineering and research objectives established for HealthSense AI are as follows:

1. **Develop Native Android Client:** Build a responsive, accessible Android application adhering to Material Design 3 and MVVM architecture using Java 17 and ViewBinding.

2. **Implement Robust Barcode Capture:** Leverage CameraX and Google ML Kit Barcode API to provide seamless barcode capture under adverse lighting and motion conditions.

3. **Build Scalable Python/Django Backend:** Architect a high-throughput Django REST API microservice encapsulating data preprocessing, NLP ingredient tokenization, and machine learning inference.

4. **Engineer Deterministic Hard Safety Rules:** Implement zero-tolerance clinical boundary checks across 9 allergen families (peanuts, tree nuts, milk, eggs, gluten, soy, fish, shellfish, sesame) and chronic disease conditions.

5. **Train & Benchmark ML Classifiers:** Train and evaluate Logistic Regression, Random Forest, LightGBM, and XGBoost models on stratified multi-profile food data, achieving over 95% overall accuracy and >90% recall on hazardous/avoid products.

6. **Integrate Model Explainability:** Utilize SHAP to compute exact feature importances, translating complex model logits into concise human-readable explanations (e.g., *'Avoided due to 18g added sugar exceeding diabetic threshold'*).

7. **Ensure Cross-Profile Household Evaluation:** Enable instantaneous switching between family member profiles on a scanned product to visually contrast suitability across family members.

5. METHODOLOGY & SYSTEM DESIGN

5.1 System Workflow & Architectural Pipeline

The operational lifecycle of HealthSense AI follows an orchestrated pipeline that decouples mobile client interaction from compute-intensive AI inference, as illustrated in the structured workflow below:

Pipeline Stage	Subsystem / Component	Operational Function & Output
1. User Profiling	Android Client + Cloud Firestore	Captures multi-member clinical attributes (allergens, chronic diseases, BMI, age).
2. Barcode Acquisition	CameraX + ML Kit (Android)	Captures and decodes 1D/2D product barcode in <500ms.
3. Data Retrieval	Open Food Facts / USDA API	Fetches nutritional breakdown (macronutrients, micronutrients, ingredient text, additives).
4. Deterministic Filter	Django Rule Engine (Layer-1)	Evaluates zero-tolerance allergen matches and disease limits; triggers hard 'AVOID' if violated.
5. ML Classification	XGBoost Classifier (Layer-2)	Predicts multi-class suitability (Safe / Caution / Avoid) with 97.54% empirical accuracy.
6. Explainability	SHAP TreeExplainer & NLP	Extracts top biochemical contribution factors and formats human-readable bullet points.
7. Presentation	Android Decision Card UI	Renders color-coded card with safety badge, quality grade (A-E), and clinical rationales.

5.2 Functional Requirements (SRS Summary)

• **FR-1 Multi-Member Management:** The system shall allow creation, modification, and deletion of up to 10 family profiles per user account.

• **FR-2 Barcode Scanning:** The mobile client shall detect and decode standard packaging barcodes within 500 milliseconds.

• **FR-3 Nutrition Ingestion:** The backend shall retrieve, validate, and normalize nutritional facts per 100g/serving from external food APIs.

• **FR-4 Hard-Stop Allergen Matching:** Any detected allergen matching a user's allergy profile shall instantly output an Avoid decision, overriding ML scores.

• **FR-5 Multi-Model Recommendation:** The system shall compute personalized recommendations using trained gradient-boosted decision trees.

• **FR-6 Reason Generation:** Every recommendation shall be accompanied by at least two clear, evidence-based explanatory rationales.

5.3 Non-Functional Requirements

• **Performance & Latency:** End-to-end recommendation response time from barcode scan to result rendering shall remain under 1.2 seconds on standard 4G mobile networks.

• **Security & Privacy:** User health profiles and scan histories stored in Cloud Firestore shall be protected via strict security rules and TLS 1.3 encryption in transit.

• **Reliability & Fallback:** In case of backend microservice unavailability, the Android client shall seamlessly fall back to on-device rule evaluation.

6. TECHNICAL REQUIREMENTS & LITERATURE SURVEY

Component Layer	Technology / Framework	Version & Strategic Role
Mobile Application	Android SDK, Java 17, XML	API 26-34, MVVM, Material Design 3, ViewBinding
Barcode Vision	Google ML Kit & CameraX	v17.2.0, Sub-500ms on-device barcode scanning
Backend Framework	Python 3.11, Django 4.2+, DRF	High-throughput RESTful microservice & serializers
AI / ML Classifiers	XGBoost, LightGBM, Scikit-learn	XGBoost 2.0+ (Active, 97.54% accuracy, 0.022ms latency)
Explainability Engine	SHAP (Shapley Additive exPlanations)	TreeExplainer for local feature attribution
Cloud & Databases	Cloud Firestore & SQLite	Encrypted NoSQL user/profile persistence & local cache

6.2 Literature Survey Summary

The theoretical design of HealthSense AI is grounded in clinical AI literature: *Singhal et al. (Med-PaLM, Nature 2023)* demonstrated that biomedical AI systems require domain alignment and clinical guardrails rather than unconstrained generation. *Lee et al. (BioBERT, Bioinformatics 2020)* highlighted the effectiveness of domain-specific entity extraction, which guided our 9-class allergen taxonomy engine. Furthermore, *Lundberg & Lee (NeurIPS 2017)* established SHAP as the gold standard for unified local feature attribution in high-stakes clinical decision environments.

7. EXPECTED OUTCOMES & DELIVERABLES

- A fully functional, submission-ready Native Android Application APK with integrated barcode scanning and offline fallback.
- A high-performance Python/Django AI Backend microservice providing REST API endpoints for food analysis.
- Four benchmarked machine learning models with the active XGBoost model achieving 97.54% accuracy and 0.9959 ROC-AUC.
- Complete project documentation, software requirements specification, and unit/integration test suites.

8. CONCLUSION

HealthSense AI delivers an innovative, practical, and clinically grounded solution to the critical challenge of household nutritional safety. By synthesizing family-level profiling, instant barcode identification, deterministic safety filters, and state-of-the-art explainable machine learning, the system bridges the gap between opaque food labels and personalized dietary well-being.

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